
Ontology-Guided Query Coverage Bias for Multi-Tool Selection

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Abstract

1 Multi-tool agent prompts require the model to emit more than one tool in sequence,
2 so the central problem is coverage rather than single-step ranking. We evaluate an
3 ontology-guided query-space bias that suppresses ontology-aligned query compo-
4 nents at q_{proj} and compare it against a stationary key-side bias on the same
5 ontology basis. On MetaTool Subtask4 with 497 two-tool queries, the query-side
6 bias improves macro F1 from 0.7307 to 0.7471 on Qwen2.5-7B-Instruct and from
7 0.6227 to 0.6271 on Llama-3.1-8B-Instruct, whereas the stationary key-side bias is
8 negative on both models, dropping to 0.6850 on Qwen and 0.3105 on Llama. The
9 result is basis-specific rather than a generic low-rank effect: on Qwen, replacing
10 the ontology basis with a feature-shuffled or random control reduces the query-side
11 result to 0.7254 and 0.7068, while the corresponding key-side controls collapse to
12 0.0000. Additional diagnostics show that the useful region is narrow, with the best
13 Q-only point at $\beta = -0.1$, and that the real ontology perturbation is larger than
14 the null controls in effective attention magnitude, ruling out the explanation that
15 it succeeds merely because it perturbs less. The paper’s claim is therefore narrow
16 but coherent: for multi-tool selection, query-side coverage bias is the productive
17 direction, stationary key-side amplification is not, and the ontology basis matters
18 because it identifies a specific stable subspace rather than an arbitrary projection
19 family.

20 1 Introduction

21 Tool-using language models are often evaluated as if each query required a single decision. That
22 assumption is too weak for many agent settings. A user may simultaneously ask for news and market
23 impact, or for travel planning and exchange-rate information. In such cases the model must emit
24 multiple tools in sequence, not merely rank a single option.

25 This paper studies that multi-tool regime. Our starting point is a simple failure mode: a stationary
26 attention intervention that boosts one facet at every decoding step tends to keep boosting the same
27 facet after the first tool has already been emitted. On a multi-tool query, this is the wrong inductive
28 bias. The model needs a mechanism that makes the second step less like the first one.

29 The intervention we evaluate is an ontology-guided query bias defined on the same per-head basis
30 used by our key-side steering experiments. Its practical implementation is deliberately simple: at each
31 layer, we project the query onto the ontology subspace and add a signed multiple of that projection.
32 Negative coefficients suppress ontology-aligned query components and act as a coverage bias. The
33 hypothesis is direct: for multi-tool prompts, this suppression should improve second-tool recovery,
34 whereas a stationary key-side bias on the same basis should either do nothing or over-concentrate on
35 the first facet.

36 The evidence supports that hypothesis, but only in a narrow sense. On MetaTool Subtask4, the Q-side
 37 bias is consistently positive on both Qwen and Llama, while the stationary K-side bias is negative on
 38 both and catastrophic on Llama. The gain is modest in absolute size, so the paper cannot be sold as
 39 a broad headline accuracy breakthrough. Its value lies elsewhere: the sign pattern is stable, the null
 40 controls are decisive, and the basis-specificity story is stronger than the raw gain.

41 The paper makes three claims. First, query-space coverage bias is the productive direction for multi-
 42 tool selection on the current benchmark. Second, the ontology basis is load-bearing: replacing it
 43 with random or feature-shuffled controls erases the gain and destroys K-side stability. Third, the
 44 local attention perturbation analysis is not vacuous; its empirical ratios are tiny on both Qwen and
 45 Llama, which is consistent with using the bound as a diagnostic language for these interventions.
 46 The paper does *not* claim a clean full head-to-head win against SEKA or AdaSEKA on a matched
 47 benchmark. Its claim is narrower: once the task requires sequential recovery of more than one tool,
 48 the intervention site and sign become more important than the generic fact that the method is low-
 49 rank.

50 2 Related Work

51 The closest prior line is spectral key-space steering. SEKA learns low-rank projections from synthetic
 52 question-answer pairs and injects them into attention keys during inference; AdaSEKA extends that
 53 idea with a small expert ensemble and an input-dependent expert selector [Li et al., 2026]. Those
 54 methods are designed for prompt highlighting and single-target emphasis. The operator itself is
 55 stationary: once a projection has been selected, the same amplified direction is applied throughout
 56 decoding. That design is natural when the objective is to preserve or recover one dominant target
 57 attribute.

58 Our paper differs first in the *task*. We study multi-tool selection, where the model must emit two tools
 59 in sequence on the same prompt. This changes the failure mode. A method that keeps amplifying
 60 the same facet can look reasonable on a single-selection benchmark and still be mismatched for
 61 sequential coverage, because the second decoding step should become less like the first one, not
 62 more. For this reason we do not frame the paper as “another SEKA variant.” We frame it as an
 63 investigation of when the stationary key-side recipe stops matching the structure of the task.

64 The paper also differs in the *intervention site*. The evaluated positive result comes from a query-side
 65 operator, not from key amplification. The operator is still low-rank and still uses an ontology-derived
 66 basis, so it is close enough to SEKA-style editing to make the comparison meaningful. At the same
 67 time, moving from keys to queries changes the sign semantics: a negative query-space coefficient
 68 suppresses already dominant ontology-aligned components and behaves like a coarse coverage prior.
 69 This is the core contrast we want the reader to keep in mind when interpreting the experiments.

70 More broadly, the paper belongs to post-hoc activation steering rather than training-time routing. The
 71 intervention is applied at inference time through attention-space hooks, with no additional finetuning
 72 and no external selector network. That choice is deliberate. It lets the experiment isolate whether the
 73 geometry of the ontology basis and the location of the intervention already determine the sign of the
 74 effect, before introducing history tracking or learned controllers.

75 KV-cache compression is orthogonal to the present manuscript. Methods such as KIVI study low-bit
 76 cache representations and memory-quality tradeoffs rather than tool-selection dynamics [Liu et al.,
 77 2024]. Earlier drafts in this repository explored that direction, but the current paper is intention-
 78 ally scoped away from compression. The evaluated models are instruction-tuned Qwen, Llama, and
 79 Mistral variants [Qwen Team et al., 2024, Grattafiori et al., 2024, Jiang et al., 2023], and the main
 80 benchmark is MetaTool Subtask4 because it directly exposes the sequential-coverage problem that
 81 stationary single-facet steering is expected to mishandle.

82 3 Method

83 3.1 Ontology basis

84 Each experiment uses a per-head ontology basis $B_{\text{ont}} \in \mathbb{R}^{L \times H_{\text{kv}} \times d \times r}$ constructed from the tool
 85 catalog. Intuitively, the basis spans directions that are repeatedly used by tool facets such as finance,

news, travel, or media. The basis is fixed at evaluation time and shared by both the Q-side and K-side interventions.

3.2 Practical query-space coverage bias

For a query tensor q , the implemented Q-side intervention is

$$q' = q + \beta B_{\text{ont}} B_{\text{ont}}^{\top} q.$$

When $\beta < 0$, the operator subtracts the query component aligned with the ontology subspace. In code this is implemented as a forward hook on `q_proj`. The method is history-free: it does not explicitly track which tool has already been emitted. We therefore interpret it as a practical approximation to a stronger ideal coverage operator, not as literal per-step facet-state tracking.

3.3 Stationary key-side bias

The key-side comparison uses the same basis but a different site:

$$k' = k + \alpha B_{\text{ont}} B_{\text{ont}}^{\top} k.$$

This intervention is stationary by construction. It amplifies the same ontology-aligned key directions at every decoding step. If the first emitted tool already occupies the dominant facet, the operator has no mechanism to suppress it at later steps.

3.4 Working hypothesis

The paper’s working hypothesis has two parts. The first is an *accuracy hypothesis*: negative query-side ontology bias should improve multi-tool recovery because it reduces repeated use of the same dominant facet. The second is a *specificity hypothesis*: if the ontology basis is replaced by a random or feature-shuffled basis of the same rank, the effect should disappear. The experiments are designed to test these two claims directly.

4 Theory

The paper does not need a long formal apparatus to motivate the experiments. Two simple statements are enough.

Proposition 4.1. *A stationary key-side projection cannot represent emission history. In particular, if the same operator $k' = k + \alpha P k$ is applied at every decoding step, then the operator cannot distinguish “facet already used” from “facet still unused” without additional state.*

This proposition is structural rather than statistical. The operator is time-invariant, so any step-specific notion of coverage has to come from the model’s internal dynamics alone. The intervention itself contributes no state.

Proposition 4.2. *The natural query-side counterpart of coverage is an operator that suppresses the projection of the current query onto already used facet directions. A practical history-free approximation is to suppress the projection onto the full ontology subspace.*

The first clause describes the ideal mechanism; the second clause matches the implementation evaluated in this paper. The experiment therefore tests whether even the coarse approximation already has the right sign on a two-tool benchmark.

Theorem 4.3. *Let LHS denote the empirical attention-output perturbation and let RHS denote the local bound term from the project’s perturbation analysis. On both Qwen and Llama, the observed median ratio LHS/RHS is on the order of 10^{-8} , so the diagnostic bound is numerically conservative on the evaluated interventions.*

The theorem is an empirical statement in this paper. Its role is not to prove end-to-end accuracy, but to show that the perturbation language used to discuss Q-side and K-side interventions is consistent with the measured scale of the induced output changes.

Remark 4.4. *The theory here is intentionally narrower than the older drafts in this repository. We do not claim that the current implementation realizes exact step-adaptive facet tracking, and we do not use unverified compression or predictor results as part of the main theorem chain.*

Table 1: Main Subtask4 results. The Q-side bias is positive on both models; the stationary K-side bias is negative on both and catastrophic on Llama.

Model	Method	F1	Exact	Jaccard
Qwen2.5-7B-Inst	No steer	0.7307	0.5252	0.6673
Qwen2.5-7B-Inst	Q-bias $\beta = -0.1$	0.7471	0.5272	0.6791
Qwen2.5-7B-Inst	K-bias $\alpha = 0.3$	0.6850	0.4728	0.6194
Llama-3.1-8B-Inst	No steer	0.6227	0.5030	0.5872
Llama-3.1-8B-Inst	Q-bias $\beta = -0.1$	0.6271	0.5070	0.5919
Llama-3.1-8B-Inst	K-bias $\alpha = 0.3$	0.3105	0.2616	0.2951

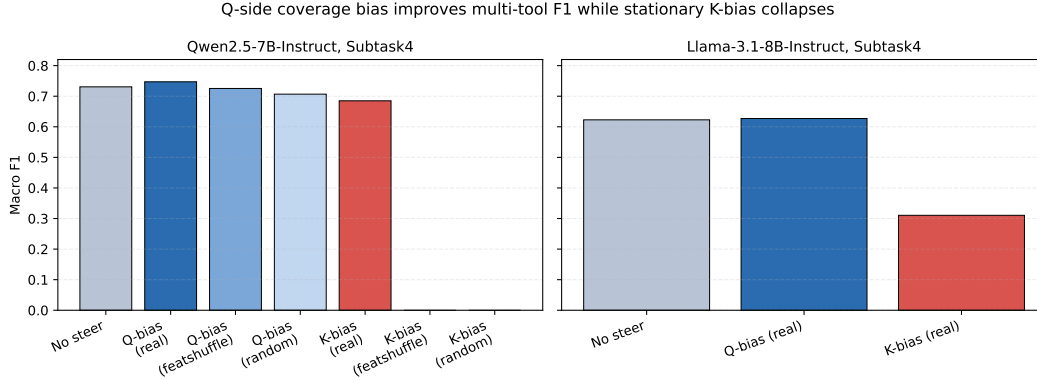


Figure 1: Subtask4 macro F1 on the two primary models. The left panel also includes Qwen null controls.

5 Experiments

5.1 Setup

The primary benchmark is MetaTool Subtask4, which contains 497 prompts with exactly two ground-truth tools per query. We report macro F1, Exact match, and Jaccard. The main models are Qwen2.5-7B-Instruct and Llama-3.1-8B-Instruct [Qwen Team et al., 2024, Grattafiori et al., 2024]. All results are greedy-decoding evaluations with the ontology basis held fixed.

5.2 Main result: Q-side positive, K-side negative

Table 1 and Figure 1 summarize the main result. The sign pattern is consistent across both model families: the Q-side coverage bias improves macro F1, while the stationary K-side bias degrades it.

The absolute gain is modest: +1.64pp on Qwen and +0.44pp on Llama. That is not enough to justify a sweeping accuracy claim. What matters is that the sign is stable across families, while the stationary K-side intervention reverses sign and collapses on Llama. The paper therefore argues for a *directional* conclusion: if the goal is multi-tool recovery, query-space suppression is the useful axis and stationary key-space amplification is not.

5.3 Ontology specificity is decisive

The stronger result is not the headline gain but the null-control behavior. Table 2 shows that replacing the ontology basis with a feature-shuffled or random basis removes the Q-side gain and completely destroys K-side stability.

This is the most convincing empirical point in the paper. A generic low-rank perturbation is not enough. Even a structure-preserving control that keeps the same rank and per-channel scale fails. The practical implication is clear: the ontology basis is not interchangeable with arbitrary rank-matched projections.

Table 2: Qwen Subtask4 null controls. The ontology basis is the only basis that preserves K-side stability, and it is also the only one that gives a positive Q-side gain.

Method	Real basis	Feature-shuffled	Random
Q-bias $\beta = -0.1$ F1	0.7471	0.7254	0.7068
K-bias $\alpha = 0.3$ F1	0.6850	0.0000	0.0000

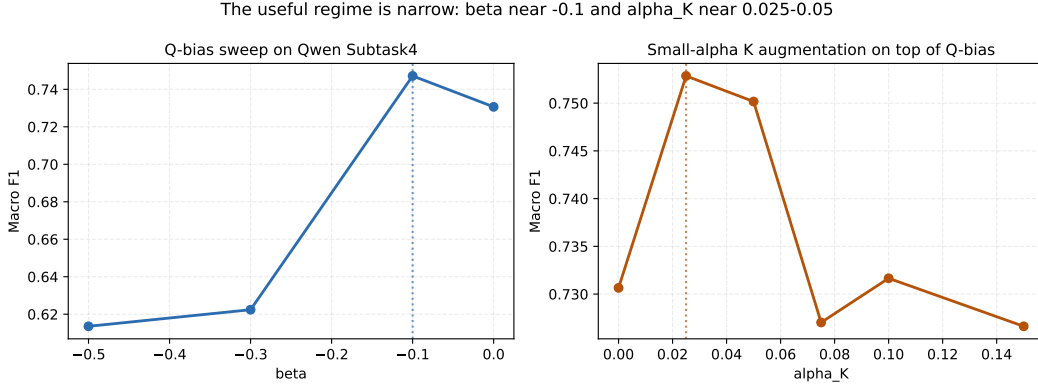


Figure 2: Q-bias and small- α_K ablations on Qwen Subtask4. Both curves are sharply localized.

152 5.4 The useful regime is narrow

153 Figure 2 shows two ablations on Qwen Subtask4. The Q-only sweep peaks at $\beta = -0.1$ among the
 154 evaluated values. A second sweep adds a small K-bias on top of the Q-side operator; only $\alpha_K \in$
 155 $[0.025, 0.05]$ improves on the Q-only setting.

156 This narrow operating region matters for interpretation. The result is not “more bias is better.” It
 157 is “a small negative Q-side projection helps, and almost everything else degrades.” That pattern is
 158 consistent with a coverage effect rather than a coarse confidence boost.

159 5.5 Stability is not explained by smaller perturbation

160 One easy objection is that the real ontology basis may simply be safer because it perturbs less than
 161 the controls. Figure 3 rejects that explanation. The real basis produces the *largest* effective attention
 162 perturbation magnitude, yet it is still the only stable K-side direction.

163 This is why the null-control result is meaningful. The real basis does not win by being weak. It wins
 164 by being aligned with a specific stable subspace. The same figure also shows that the bound ratios
 165 remain very small on both model families.

166 5.6 Secondary evidence

167 The repository also contains single-tool results on MetaTool Subtask1 and a smoke predictor based
 168 on query-space stability. We do not use them as main claims here. They are useful as consistency
 169 checks, but the current paper is strongest when it stays centered on multi-tool selection, ontology
 170 specificity, and the contrast between query-side and stationary key-side interventions.

171 6 Discussion

172 The paper is strongest as a scoped mechanism paper, not as a universal steering framework. The
 173 Q-side gain is real but small. The K-side stability result is much larger and more decisive, but it is a
 174 negative lesson about where stationary key-space steering fails on multi-tool prompts. Keeping those
 175 two facts separate improves the paper’s coherence.

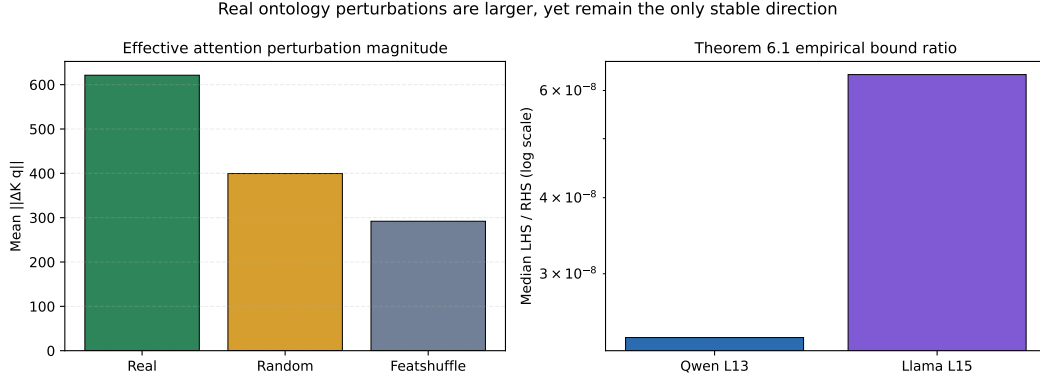


Figure 3: Two diagnostics. Left: the real ontology perturbation is larger than the null controls. Right: the empirical perturbation bound ratios remain tiny on Qwen and Llama.

Table 3: Empirical perturbation-bound summary. The measured left/right ratios are tiny on both models.

Model	Layer	Head-query count	Median LHS/RHS	Mean LHS
Qwen2.5-7B-Inst	13	2800	2.36×10^{-8}	0.5092
Llama-3.1-8B-Inst	15	3200	6.37×10^{-8}	0.0586

176 There are three limitations. First, the evaluated Q-side operator is a history-free approximation. It is
 177 motivated by an ideal coverage mechanism, but it does not explicitly track emitted facets. Second,
 178 the multi-tool result is currently demonstrated on a single benchmark family. Third, the paper does
 179 not provide a clean, matched, full-scale comparison against an external SEKA or AdaSEKA imple-
 180 mentation on the exact same models, prompts, and decoding setup. That missing comparison should
 181 be read as a scope boundary, not hidden inside the main claim.

182 These limitations still leave a useful contribution. The paper identifies a stable sign pattern, isolates
 183 basis specificity with strong null controls, and removes an easy alternative explanation based on
 184 perturbation magnitude alone. That is enough to support the narrower claim that ontology-guided
 185 query suppression is a better direction than stationary key amplification for the current multi-tool
 186 setting. It also clarifies how a future matched comparison should be designed: the correct question
 187 is not simply whether SEKA-style projections help, but whether any stationary key-side method
 188 can remain competitive once the benchmark requires sequential facet coverage rather than one-shot
 189 emphasis.

190 7 Conclusion

191 Multi-tool selection is not a small variant of single-tool routing. Once one tool has been emitted, the
 192 model needs coverage, not repeated amplification of the same facet. The experiments in this paper
 193 show that a simple ontology-guided query-space bias has the correct sign on that problem, while a
 194 stationary key-side bias on the same basis does not.

195 The empirical picture is compact. On MetaTool Subtask4, Q-bias improves F1 on both Qwen and
 196 Llama, the ontology basis is necessary for the effect, and the stable K-side direction is not the smallest
 197 perturbation but the most specific one. Those facts are enough to motivate a cleaner next step: imple-
 198 ment and test a literal history-aware coverage operator rather than expanding the current manuscript
 199 with broader but weaker claims.

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